

Ivan Romero Moreno

Software Engineer

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Software Engineer with 3 years of experience building scalable backend systems and microservices in Python. Experienced in designing event-driven architectures, deploying production services on AWS, and building data-intensive ML pipelines. Proven track record delivering reliable systems for large-scale document processing workloads (100k+ daily documents) in the banking sector, with end-to-end ownership from architecture to production deployment.

SKILLS

Programming Languages: Python, SQL, Bash, TypeScript, Git

Frameworks & Cloud: FastAPI, Django, React, PostgreSQL, Redis, Docker, AWS, Nginx, Linux, CI/CD, PyTest

ML & Data Science: PyTorch, CNNs, NLP/NER, LLMs, Agentic AI, LangChain

Languages: English (fluent), Spanish (native), Catalan (native)

WORK EXPERIENCE

Freelance Full-Stack Engineer

Sept 2025 – Jan 2026

Self-employed – Remote

- Built a Business Intelligence platform for White Rabbit Museum (client: Rivex) with FastAPI, PostgreSQL, Redis, React/TypeScript and a LangChain-based RAG chatbot to enable natural-language exploration of visitor analytics.
- Delivered an industrial maintenance platform for Engi-On, eliminating 100% of paper-based maintenance workflows by designing and deploying a Django REST backend with a React/TypeScript frontend.
- Containerized and deployed both applications on AWS, improving reliability with CI/CD pipelines using GitHub Actions, Nginx reverse proxy, and Prometheus/Grafana.

Software Engineer

Feb 2024 – Aug 2025

Serimag – Barcelona, Spain

- Mentored and led a team of 3 engineers on Django/FastAPI/REST API development, delivering 5+ production microservices for a major Spanish banking client.
- Evolved Python orchestration platform using event-driven architecture to coordinate business logic across 10+ microservices (including data-intensive ML pipelines), reliably processing 100,000+ documents daily.
- Designed and deployed a High Availability MySQL cluster with automatic failover for a critical authentication service, reducing recovery time from hours to under one minute.
- Owned the end-to-end deployment lifecycle with weekly microservice releases on Linux using Docker, Nginx, CI/CD pipelines and automated testing with PyTest.

AI Engineer Intern

Jan 2023 – Feb 2024

Serimag – Barcelona, Spain

- Led the development of real-time CNN-based document classification models processing 30,000+ documents daily for a major Spanish bank, reducing manual document handling by 80%.
- Engineered NLP and NER pipelines that increased automated invoice information extraction from 60% to 70% at 99% accuracy, saving 35 manual hours per week across workflows processing 5,000+ documents daily.
- Audited and improved production ML models in banking workflows, maintaining misclassification rates below 1% through model evaluation, dataset refinement, and performance monitoring.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate

Mar 2026

Amazon Web Services

EDUCATION

Polytechnic University of Catalonia – Barcelona, Spain

Sept 2024 – Jun 2026 (expected)

Master's in Advanced Telecommunications Technologies (Specialization in Deep Learning for Multimedia Processing)

Polytechnic University of Catalonia – Barcelona, Spain

Sept 2018 – Jan 2024

Bachelor's in Telecommunications Engineering